

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
##     stamp
```

```
library(ggthemes)
```

```
##  
## Attaching package: 'ggthemes'  
##  
## The following object is masked from 'package:cowplot':  
##  
##     theme_map
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
PeterPaul_ChemPhys <-  
  read_csv(here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"))
```

```
## Rows: 23008 Columns: 15  
## -- Column specification -----  
## Delimiter: ","  
## chr   (1): lakename  
## dbl  (13): year4, daynum, month, depth, temperature_C, dissolvedOxygen, irra...  
## date  (1): sampledate  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
PeterPaul_Litter <-  
  read_csv(here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"))
```

```
## Rows: 1692 Columns: 13  
## -- Column specification -----  
## Delimiter: ","  
## chr   (7): plotID, trapID, functionalGroup, qaDryMass, nlcdClass, plotType, g...  
## dbl   (5): dryMass, subplotID, decimalLatitude, decimalLongitude, elevation  
## date  (1): collectDate  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#2
glimpse(PeterPaul_ChemPhys$sampldate)

## Date[1:23008], format: "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" ...

glimpse(PeterPaul_Litter$collectDate)

## Date[1:1692], format: "2016-06-16" "2016-06-16" "2016-06-16" "2016-06-16" "2016-06-16" ...
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
myTheme1 <- theme_base() +
  theme(
    plot.background = element_rect(
      color = 'black',
      fill = 'white'
    ),
    plot.title = element_text(
      color = 'black'
    ),
    axis.text = element_text(
      color = 'black'
    ),
    axis.ticks = element_line(
      color = 'purple',
      linewidth = 1
    )
  )
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4
PvsP04_plot <- PeterPaul_ChemPhys %>%
  ggplot(aes(
    x=po4,
    y=tp_ug,
    color=lakename)
  ) +
  geom_point() +
  labs(x = "Phosphate (po4)",
       y = "Total Phosphorus (tp_ug)",
       title = "Total Phosphorus by Phosphate",
       color = "Lake") +
  geom_smooth(method = "lm", se = FALSE, size = 1) +
  xlim(0,40) +
  ylim(0,125)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
facet_wrap(facet=vars(lakename),
           nrow=2)
```

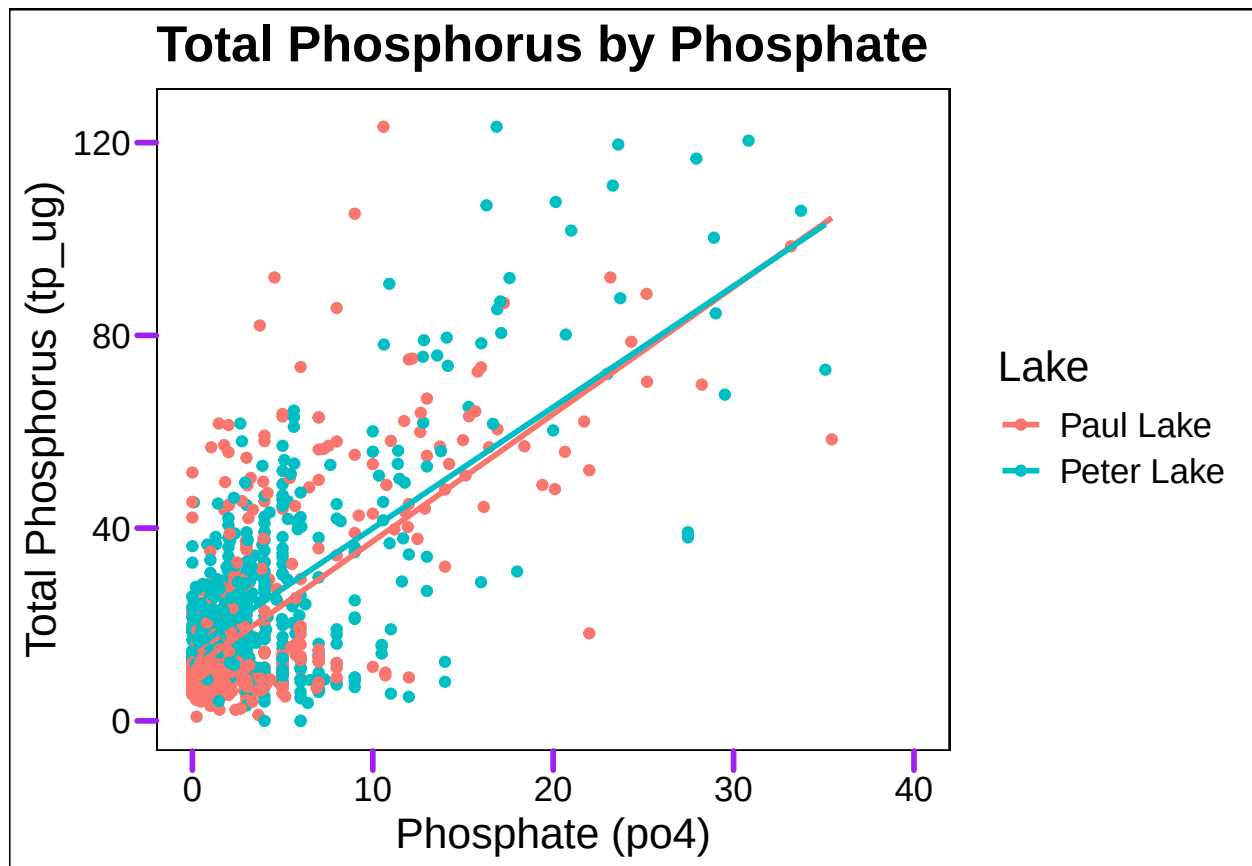
```
## <ggproto object: Class FacetWrap, Facet, gg>
##   compute_layout: function
##   draw_back: function
##   draw_front: function
##   draw_labels: function
##   draw_panels: function
##   finish_data: function
##   init_scales: function
##   map_data: function
##   params: list
##   setup_data: function
##   setup_params: function
##   shrink: TRUE
##   train_scales: function
##   vars: function
##   super: <ggproto object: Class FacetWrap, Facet, gg>
```

```
PvsP04_plot + myTheme1
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21953 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21953 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
NTLTempBox <- PeterPaul_ChemPhys %>%
  ggplot(aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = temperature_C,
    color = lakename)
  ) +
  geom_boxplot() +
  labs(x = "Month",
       y = "Temperature (C)",
       color = "Lake") +
  theme(legend.position = "none",
        axis.title.x = element_blank())

NTLTPBox <- PeterPaul_ChemPhys %>%
  ggplot(aes(
```

```

    x = factor(month, levels = 1:12, labels = month.abb),
    y = tp_ug,
    color = lakename)
  ) +
  geom_boxplot() +
  labs(x = "Month",
       y = "TP",
       color = "Lake") +
  theme(axis.title.x = element_blank())

NTLTNBox <- PeterPaul_ChemPhys %>%
  ggplot(aes(
    x = factor(month, levels = 1:12, labels = month.abb),
    y = tn_ug,
    color = lakename)
  ) +
  geom_boxplot() +
  labs(x = "Month",
       y = "TN",
       color = "Lake") +
  theme(legend.position = "none")

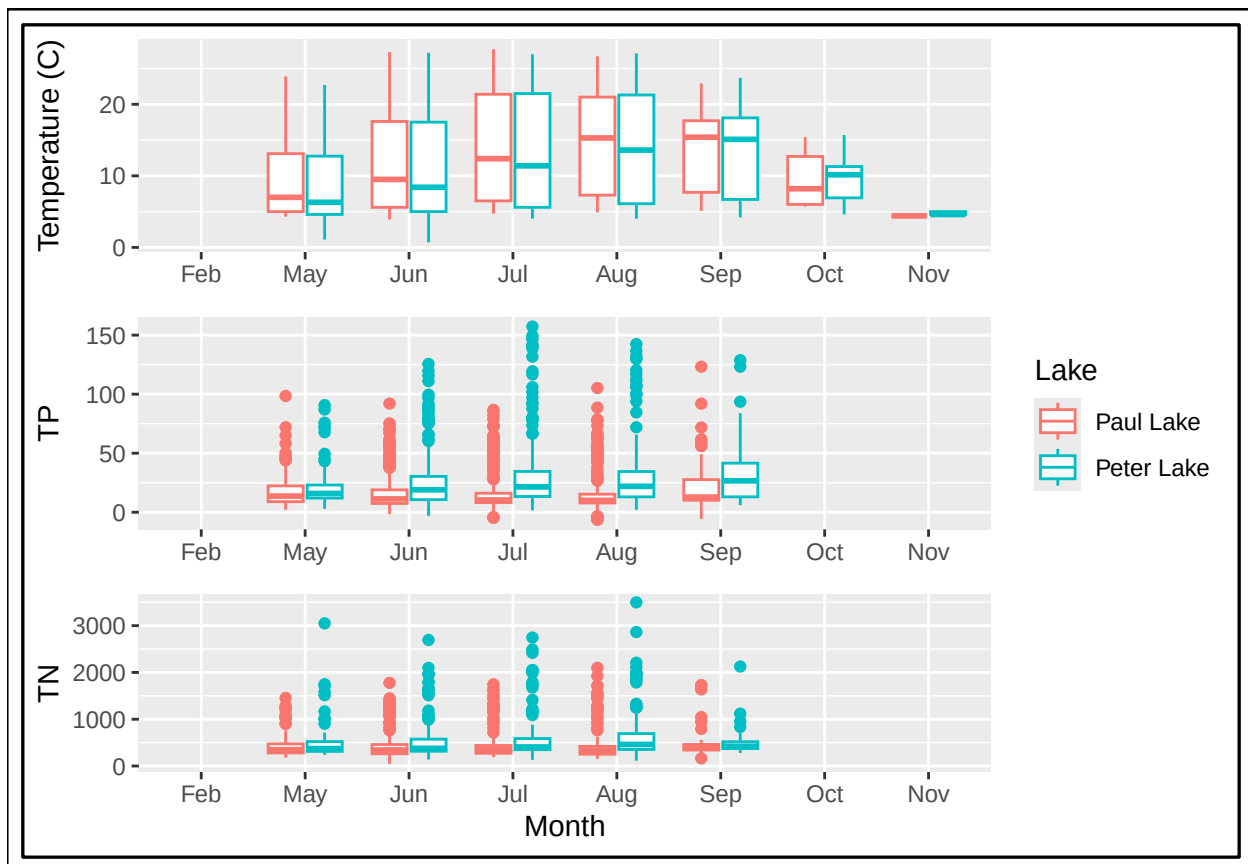
plot_grid(NTLTempBox, NLTTPBox, NTLTNBox, nrow = 3, align = "v") + myTheme1

## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

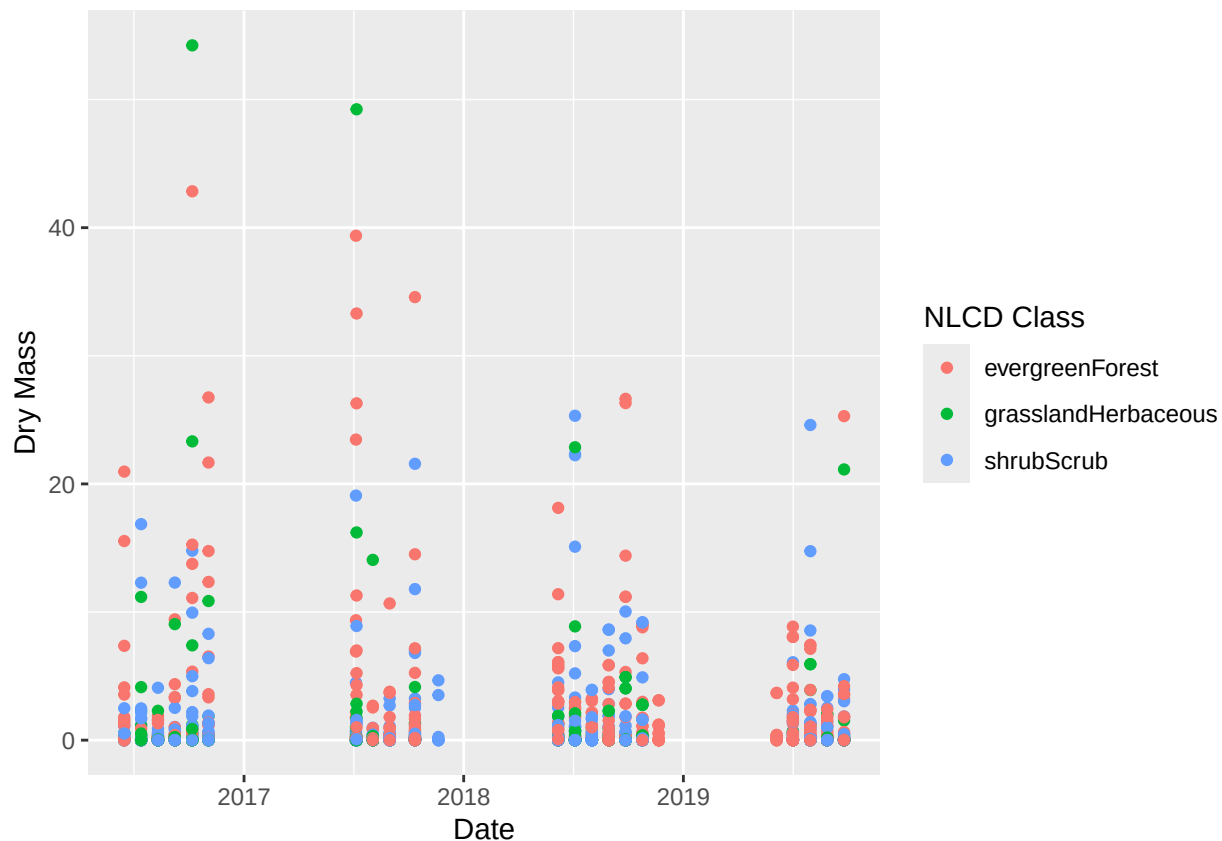


Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: We observe that temperature, TP, and TN levels all are at their highest in the summertime, specifically in July and August. It seems that Peter Lake has higher levels of TP and TN than Paul Lake, especially TP where as the year progresses we see TP levels as we get to September. Temperature is the only variable of interest that stays about even through out the year for Peter and Paul Lakes.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
Needle_Plot <- PeterPaul_Litter %>%
  filter(functionalGroup == "Needles")
ggplot(data = PeterPaul_Litter, aes(
  x = collectDate,
  y = dryMass,
  color = nlcdClass)
) +
  geom_point() +
  labs(x = "Date",
       y = "Dry Mass",
       color = "NLCD Class")
```



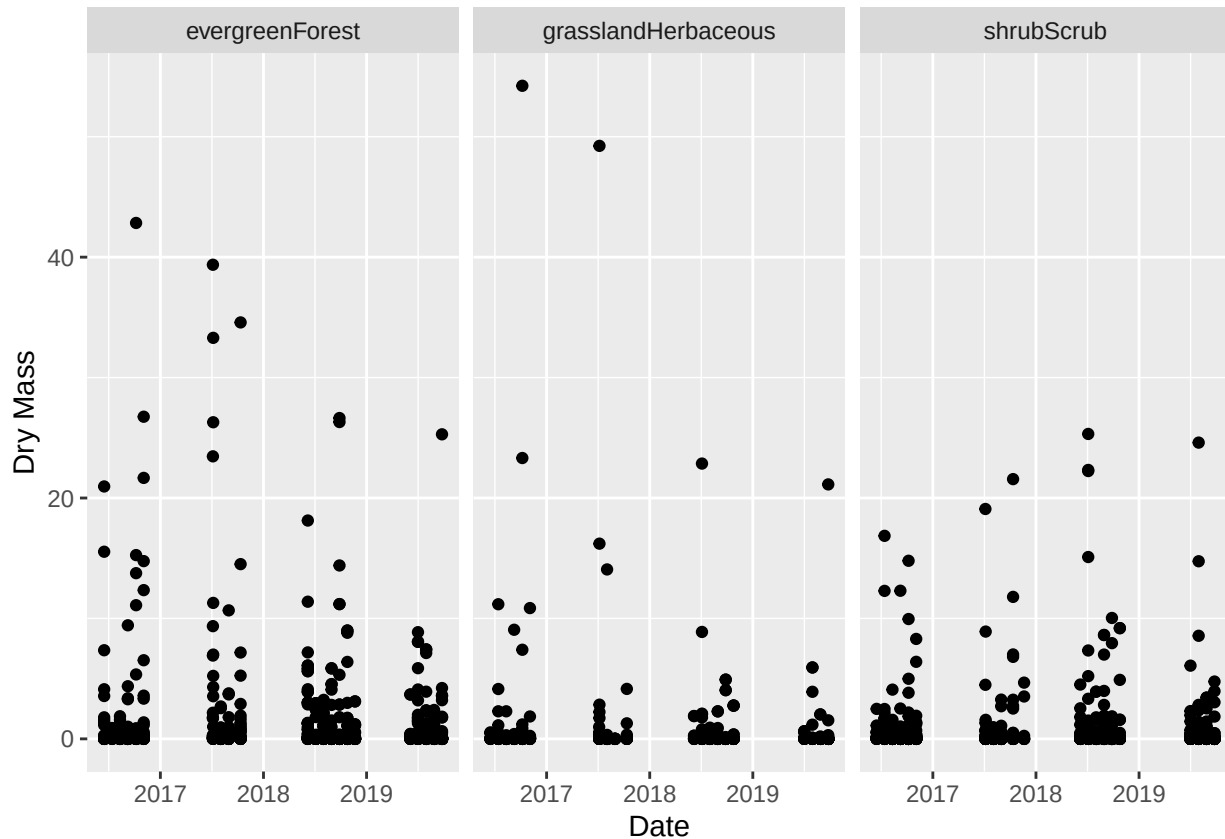
Needle_Plot

```
## # A tibble: 241 x 13
##   plotID   trapID   collectDate functionalGroup dryMass qaDryMass subplotID
##   <chr>    <chr>    <date>      <chr>                <dbl> <chr>      <dbl>
## 1 NIWO_058 NIWO_058_101 2016-06-16 Needles                1.11 Y        32
## 2 NIWO_047 NIWO_047_197 2016-06-16 Needles                0.54 N        40
## 3 NIWO_057 NIWO_057_081 2016-06-16 Needles               21.0  N        31
## 4 NIWO_064 NIWO_064_103 2016-06-16 Needles                3.56 N        32
## 5 NIWO_061 NIWO_061_169 2016-06-16 Needles               15.5  Y        41
## 6 NIWO_062 NIWO_062_050 2016-06-16 Needles                2.49 N        31
## 7 NIWO_046 NIWO_046_155 2016-07-14 Needles               16.9  Y        40
## 8 NIWO_051 NIWO_051_045 2016-07-14 Needles                4.14 Y        31
## 9 NIWO_047 NIWO_047_197 2016-07-14 Needles                0.31 Y        40
## 10 NIWO_061 NIWO_061_169 2016-07-14 Needles                0.9  Y        41
## # i 231 more rows
## # i 6 more variables: decimalLatitude <dbl>, decimalLongitude <dbl>,
## #   elevation <dbl>, nlcdClass <chr>, plotType <chr>, geodeticDatum <chr>
```

```
#7
Needle_Plot_Facets <- PeterPaul_Litter %>%
  filter(functionalGroup == "Needles")
ggplot(data = PeterPaul_Litter, aes(
  x = collectDate,
  y = dryMass))
```



```
) +
geom_point() +
facet_wrap(facets = vars(nlcdClass)) +
labs(x = "Date",
y = "Dry Mass",
color = "NLCD Class")
```



Needle_Plot_Facets

```
## # A tibble: 241 x 13
##   plotID trapID collectDate functionalGroup dryMass qaDryMass subplotID
##   <chr>   <chr>   <date>      <chr>          <dbl> <chr>      <dbl>
## 1 NIWO_058 NIWO_058_101 2016-06-16 Needles         1.11 Y         32
## 2 NIWO_047 NIWO_047_197 2016-06-16 Needles         0.54 N         40
## 3 NIWO_057 NIWO_057_081 2016-06-16 Needles        21.0  N         31
## 4 NIWO_064 NIWO_064_103 2016-06-16 Needles         3.56 N         32
## 5 NIWO_061 NIWO_061_169 2016-06-16 Needles        15.5  Y         41
## 6 NIWO_062 NIWO_062_050 2016-06-16 Needles         2.49 N         31
## 7 NIWO_046 NIWO_046_155 2016-07-14 Needles        16.9  Y         40
## 8 NIWO_051 NIWO_051_045 2016-07-14 Needles         4.14 Y         31
## 9 NIWO_047 NIWO_047_197 2016-07-14 Needles         0.31 Y         40
## 10 NIWO_061 NIWO_061_169 2016-07-14 Needles         0.9  Y         41
## # i 231 more rows
## # i 6 more variables: decimalLatitude <dbl>, decimalLongitude <dbl>,
## #   elevation <dbl>, nlcdClass <chr>, plotType <chr>, geodeticDatum <chr>
```

Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think that the plot where the each class is separated into facets is the more effective plot. While I usually like having all data condensed into one plot for the sake of convenience, there are too many data points to see things clearly. So having them separated into their own plots makes it way easier to interpret.